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System including fuel cell assembly voltage monitor

Abstract

A method for controlling a carbon dioxide utilization in a fuel cell assembly includes: measuring a voltage across the fuel cell assembly; determining an estimated carbon dioxide utilization of the fuel cell assembly based on at least the measured voltage across the fuel cell assembly by determining an expected voltage of the fuel cell assembly based on at least a temperature of the fuel cell assembly, a current density across the fuel cell assembly, a fuel utilization of the fuel cell assembly, and a cathode oxygen utilization of the fuel cell assembly; determining the estimated carbon dioxide utilization based on a comparison between the measured voltage and the determined expected voltage; comparing the determined estimated carbon dioxide utilization to a predetermined threshold utilization; and upon determining that the determined estimated carbon dioxide utilization is higher than the predetermined threshold utilization, reducing the carbon utilization of the fuel cell assembly.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/198,771, filed Mar. 11, 2021, which is a bypass continuation of PCT Patent Application No. PCT/IB2019/057671, filed Sep. 11, 2019, which claims priority from Provisional Patent Application No. 62/730,339, filed Sep. 12, 2018. The entire disclosures of these applications are incorporated herein by reference.

BACKGROUND

- (1) The present disclosure relates generally to the field of fuel cell power production systems. In particular, the present disclosure relates to systems and methods useable in a fuel cell power production system for controlling carbon dioxide capture and utilization.
- (2) A fuel cell, such as a molten carbonate fuel cell (MCFC), is a device which directly converts chemical energy stored in hydrocarbon fuel into electrical energy via an electrochemical reaction. Generally, a fuel cell includes an anode and a cathode separated by an electrolyte layer or matrix which conducts electrically charged ions. To increase power production or output, multiple fuel cells are arranged in series (also known as a fuel cell stack), with an electrically conductive separator between each individual cell.
- (3) As a molten carbonate fuel cell generates an electrical current, carbon dioxide is transferred from the cathode (or oxidant) side to the anode (or fuel side) of the fuel cell. At the cathode side, the carbon dioxide is dilute and accordingly more difficult to separate from other gaseous substances (such as nitrogen or oxygen). At the anode side, the carbon dioxide is more concentrated and accordingly more easily separated from the primary co-component of water and hydrogen.
- (4) When the MCFC is operated in a carbon capture mode, flue gas containing carbon dioxide from an external source (such as exhaust from a carbonaceous-fuel-fired power plant) is fed to the cathode side of the fuel cell. The fuel cell is operated to maintain a predetermined carbon capture rate, but there may be variability in the carbon content of the flue gas as it is introduced to the cathode side of the fuel cell. If carbon dioxide over-utilization occurs (such as removal of excessive amounts of carbon dioxide from the system), the fuel cell(s)' durability and/or life could be affected.
- (5) Accordingly, there is a need for systems and methods useable in a fuel cell system for preventing carbon dioxide over-utilization by controlling a carbon capture rate from the flue gas provided to a cathode of a fuel cell and ensure that a high percentage of the total available carbon

in the flue gas is captured while achieving efficient operation of the fuel cell system.

SUMMARY OF THE INVENTION

(6) According to one exemplary embodiment of the present disclosure, a system for capturing carbon dioxide in flue gas generated by a flue gas generating device includes a fuel cell assembly including at least one fuel cell including a cathode portion configured to receive, as cathode inlet gas, the flue gas generated by the flue gas generating device or a derivative thereof, and to output cathode exhaust gas, and an anode portion configured to receive an anode inlet gas and to output anode exhaust gas; a fuel cell assembly voltage monitor configured to measure a voltage across the fuel cell assembly; and a controller configured to receive the measured voltage across the fuel cell assembly from the fuel cell assembly voltage monitor, determine an estimated carbon dioxide utilization of the fuel cell assembly based on at least the measured voltage across the fuel cell assembly, and when the determined estimated carbon dioxide utilization is above a predetermined threshold utilization, reduce the carbon dioxide utilization of the fuel cell assembly.

(7) According to one aspect of the system, the controller is configured to determine the estimated carbon dioxide utilization by performing steps including determining an expected voltage across the fuel cell assembly based on at least (i) a temperature of the fuel cell assembly, (ii) a current density across the fuel cell assembly, (iii) a fuel utilization of the fuel cell assembly, and (iv) a cathode oxygen utilization of the fuel cell assembly, and determining the estimated carbon dioxide utilization based on a comparison between the measured voltage across the fuel cell assembly and the determined expected voltage across the fuel cell assembly.

(8) According to one aspect, the system also includes at least one temperature sensor configured to measure temperatures within the fuel cell assembly. The controller is configured to determine an average temperature of the fuel cell assembly based on the temperatures measured by the at least one temperature sensor.

(9) According to one aspect, the system also includes a load controller configured to measure the current across the fuel cell assembly. The controller is configured to determine the current density across the fuel cell based on at least the current across the fuel cell assembly as measured by the load controller. The load controller may then increase or decrease the measured load to affect the current density such that the fuel cell operates at a desired current density for any particular operation.

(10) According to one aspect, the system also includes a flow transmitter configured to measure a flow rate of the anode inlet gas. In particular, the flow transmitter may measure a flow rate of a fuel gas flow, where the anode inlet gas comprises a fuel gas flow. The controller is configured to determine the fuel utilization of the fuel cell assembly based on at least the flow rate of the fuel gas flow as measured by the flow transmitter.

(11) According to one aspect, the system also includes at least one valve configured to adjust an amount of anode inlet gas provided to the anode portion of the fuel cell assembly. The controller is configured to control the at least one valve and to determine the fuel utilization of the fuel cell assembly based on at least the flow rate of the anode inlet gas as measured by the flow transmitter.

(12) According to one aspect, the system also includes a hydrogen gas analyzer configured to measure a hydrogen content of the anode exhaust gas. The controller is configured to determine the fuel utilization of the fuel cell assembly based on at least the hydrogen concentration in the anode exhaust gas as measured by the hydrogen gas analyzer.

(13) According to one aspect, the system also includes a cathode inlet gas flow transmitter configured to measure a flow rate of the cathode inlet gas; and a cathode inlet gas analyzer configured to measure a composition of the cathode inlet gas. The controller is configured to determine the cathode oxygen utilization based on at least the flow rate of the cathode inlet gas as measured by the cathode inlet gas flow transmitter, and the composition of the cathode inlet gas as measured by the cathode inlet gas analyzer.

(14) According to one aspect, the system also includes a cathode outlet gas flow transmitter

configured to measure a flow rate of the cathode outlet gas; and a cathode outlet gas analyzer configured to measure a composition of the cathode outlet gas. The controller is configured to determine the cathode oxygen utilization based on at least the flow rate of the cathode outlet gas as measured by the cathode outlet gas flow transmitter, and the composition of the cathode outlet gas as measured by the cathode outlet gas analyzer.

(15) According to one aspect, the system also includes a cathode inlet gas flow transmitter configured to measure a flow rate of the cathode inlet gas; a cathode inlet gas analyzer configured to measure a composition of the cathode inlet gas, a cathode outlet gas flow transmitter configured to measure a flow rate of the cathode outlet gas; and a cathode outlet gas analyzer configured to measure a composition of the cathode outlet gas. The controller is configured to determine the cathode oxygen utilization based on at least the flow rate of the cathode inlet gas as measured by the cathode inlet gas flow transmitter, the measured composition of the cathode inlet gas as measured by the cathode inlet gas analyzer, the measured flow rate of the cathode outlet gas as measured by the cathode outlet gas flow transmitter, and the composition of the cathode outlet gas as measured by the cathode outlet gas analyzer.

(16) According to one aspect, the system also includes a flue gas blower configured to receive flue gas from the flue gas generating device and output the flue gas to the anode gas oxidizer of the fuel cell assembly. The controller is configured to, when the determined estimated carbon dioxide utilization is above a predetermined threshold utilization, reduce the carbon dioxide utilization of the fuel cell assembly by at least controlling the flue gas blower to increase the flow rate of the flue gas, or the derivative thereof, provided to the cathode portion of the fuel cell assembly.

(17) According to one aspect, the system also includes a load controller configured to control a current across the fuel cell assembly. The controller is configured to, when the determined estimated carbon dioxide utilization is above a predetermined threshold utilization, reduce the carbon dioxide utilization of the fuel cell assembly by at least controlling the load controller to reduce the current across the fuel cell assembly.

(18) According to one aspect, the system also includes an anode exhaust processor configured to receive the anode exhaust gas, provide an anode exhaust gas return portion of the anode exhaust gas to an anode exhaust gas return line that is configured to provide the anode exhaust gas return portion to the cathode portion of the fuel cell assembly that is equipped with an anode gas oxidizer in a manner that is controllable by the controller (e.g., by modulation of a control valve), provide an anode exhaust gas recycle portion of the anode exhaust gas to an anode exhaust gas recycle line that is configured to provide the anode exhaust gas recycle portion to the anode section of the fuel cell assembly in a manner that is controllable by the controller (e.g., by modulation of a control valve), provide carbon dioxide separated from the anode exhaust gas to a carbon dioxide product line in a manner that is controllable by the controller, and provide water separated from the anode exhaust gas to a water product line in a manner that is controllable by the controller.

(19) According to one aspect of the system, the controller is configured to, when the determined estimated carbon dioxide utilization is above a predetermined threshold utilization, reduce the carbon dioxide utilization of the fuel cell assembly by at least controlling the anode exhaust processor to reduce an amount of carbon dioxide product provided to the carbon dioxide product line, and increase an amount of the anode exhaust gas return portion provided to the cathode portion of the fuel cell assembly via the anode exhaust gas return line.

(20) According to one aspect of the system, the anode inlet gas is a combination of fuel gas provided via a fuel gas inlet line, and water provided via a water inlet line, and the at least one valve includes a first valve configured to adjust an amount of fuel gas provided via the fuel gas inlet line, and a second valve configured to adjust an amount of water provided via the water inlet line.

(21) According to one aspect, the system also includes an anode gas oxidizer configured to receive either one or both of the flue gas generated by the flue gas generating device and ambient air, and

may also receive the anode exhaust gas return portion, oxidize any unreacted fuel which may be present in the anode exhaust gas return portion using the flue gas generated by the flue gas generating device and ambient air, and output the oxidized mixed gas to the cathode portion of the fuel cell assembly as the cathode inlet gas.

(22) According to another exemplary embodiment of the present disclosure, a method for controlling a carbon dioxide utilization in a fuel cell assembly includes the steps of measuring a voltage across the fuel cell assembly using a fuel cell assembly voltage monitor, determining an estimated carbon dioxide utilization of the fuel cell assembly based on at least the measured voltage across the fuel cell assembly and an expected voltage of the fuel cell assembly, the expected voltage being determined based on at least a temperature of the fuel cell assembly, a current density across the fuel cell assembly, a fuel utilization of the fuel cell assembly, and a cathode oxygen utilization of the fuel cell assembly, and determining the estimated carbon dioxide utilization based on a comparison between the measured voltage across the fuel cell assembly and the determined expected voltage across the fuel cell assembly, comparing the determined estimated carbon dioxide utilization of the fuel cell assembly to a predetermined threshold utilization, and if the determined estimated carbon dioxide utilization of the fuel cell assembly is higher than the predetermined threshold utilization, reducing the carbon dioxide utilization of the fuel cell assembly.

(23) According to one aspect, the method also includes measuring a flow rate of a fuel gas flow, where the anode inlet gas comprises at least the fuel gas flow and the water flow; and determining the fuel utilization of the fuel cell assembly based on at least the measured flow rate of the fuel gas flow.

(24) According to one aspect, the method also includes measuring a hydrogen content of an anode exhaust gas; and determining the fuel utilization of the fuel cell assembly based on at least the measured hydrogen concentration in the anode exhaust gas.

(25) According to one aspect, the method also includes measuring a flow rate of a cathode inlet gas; measuring a composition of the cathode inlet gas; and determining the cathode oxygen utilization of the fuel cell assembly based on at least the measured flow rate of the cathode inlet gas and the composition of the cathode inlet gas.

(26) According to one aspect, the method also includes measuring a flow rate of a cathode outlet gas; measuring a composition of the cathode outlet gas; and determining the cathode oxygen utilization of the fuel cell assembly based on at least the measured flow rate of the cathode outlet gas and the composition of the cathode outlet gas.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a system diagram showing a fuel cell system adapted for capturing carbon dioxide in flue gas, according to an exemplary embodiment.

(2) FIG. 2 is a schematic control diagram of the fuel cell system adapted for capturing carbon dioxide in flue gas shown in FIG. 1.

(3) FIG. 3 is a graph illustrating a relationship between carbon dioxide flux of a fuel cell assembly and voltage across the fuel cell assembly, controlling for a percent of carbon dioxide captured.

(4) FIG. 4 is a flow chart illustrating a method of controlling a carbon dioxide capture rate from a fuel cell, according to an exemplary embodiment.

DETAILED DESCRIPTION

(5) Certain embodiments described herein are useable in maximizing the carbon capture rate within a fuel cell while not posing undue risks to the fuel cell caused by carbon dioxide over-utilization. Certain embodiments may also yield the beneficial result of increasing an efficiency of the delivery

of a secondary product, the electricity exported from the fuel cell. Certain embodiments described herein implement a cost-effective way of determining carbon dioxide usage. Rather than measuring carbon dioxide usage (which can be expensive), certain embodiments described herein compute carbon dioxide usage based on a comparison of the measured and expected voltage. Another beneficial result of certain embodiments may be an increase in efficiency of exported hydrogen gas from the fuel cell system as a tertiary product. By determining the carbon content of a flue gas provided to the fuel cell system, the cathode flow rate and/or fuel cell current can be modulated to yield a desired product mix of carbon sequestration, power output, and hydrogen export.

(6) FIG. 1 shows a fuel cell system 1 according to an exemplary embodiment of the present disclosure. The fuel cell system 1 includes a fuel cell assembly 50 having a cathode portion (e.g., cathode section) 51 and an anode portion (e.g., anode section) 52.

(7) The cathode portion 51 of the fuel cell assembly 50 is configured to receive a cathode inlet gas flow 53. The cathode inlet gas flow 53 includes a flue gas flow 6 from a flue gas generating device 5 and/or a derivative gas flow 15 which is derived from the flue gas flow 6. The cathode inlet gas flow 53 is provided to the cathode portion of the fuel cell assembly 50 by an anode gas oxidizer 25 (described below). The cathode 51 is also configured to output a cathode exhaust gas flow (e.g., cathode outlet gas) 54.

(8) The anode portion 52 of the fuel cell assembly 50 is configured to receive an anode inlet gas flow 35. The anode portion 52 of the fuel cell assembly 50 is also configured to output an anode exhaust gas flow (e.g., anode outlet gas) 55. According to one aspect, the anode inlet gas includes a water flow 8 provided to the anode portion 52 via a water inlet line and a fuel gas flow 30 provided to the anode portion 52 via a fuel gas inlet line. The flow of the water flow 8 is controllable by a valve 33 and a flow rate of the water flow 8 is measured by the flow transmitter 34. The flow of the fuel gas flow 30 is controllable by a valve 31, and a flow rate of the fuel gas flow 30 is measured by a flow transmitter 32. The flow transmitter 74 is configured to measure the flow rate of anode exhaust gas recycle flow as controlled by valve 73. The total anode inlet gas flow 35 is the sum of the flows 32, 34 and 74. The valves 31, 33 are configured to control an amount of the fuel gas flow 30 and an amount of the water flow 8, respectively, delivered to the anode portion 52 of the fuel cell assembly 50. The valves 31, 33 are controllable by the controller 100, shown in FIG. 2, such that when a determined estimated carbon dioxide utilization is above a predetermined threshold utilization, the controller 100 reduces the current flow in the fuel cell assembly 50 and to reduce the carbon dioxide utilization of the fuel cell assembly 50 and in turn adjusts the valves 31, 33 to decrease an amount of the anode inlet gas flow 35 provided to the anode portion 52 of the fuel cell assembly to match the anode inlet flow needed for the reduced current density. The anode portion 52 of the fuel cell assembly 50 is also configured to output an anode exhaust gas flow 55 to an anode exhaust processor 65 (described in more detail below).

(9) In the fuel cell assembly 50, the carbon dioxide from the cathode portion 51 is electrochemically pumped to the anode portion 52. Thus the carbon dioxide moves from an environment (cathode portion 51) in which the concentration of carbon dioxide is low and diluted with species of oxygen and nitrogen from which the carbon dioxide is difficult to remove to an environment (anode portion 52) in which the concentration of carbon dioxide is relatively high and in the presence of species of hydrogen and water from which the carbon dioxide is more easily removed. Accordingly, the carbon dioxide, because it is more highly concentrated and physically dissimilar from primary co-components in the anode portion 52 of the fuel cell assembly 50, can more easily be separated from the primary co-components of water and hydrogen.

(10) The fuel cell system 1 also includes a fuel cell assembly voltage monitor 90 electronically coupled to the fuel cell assembly 50. The fuel cell assembly voltage monitor 90 is configured to measure a voltage across the fuel cell assembly 50. The fuel cell assembly voltage monitor 90 is configured to send a signal to a controller 100 (described below) based on the measured voltage across the fuel cell assembly 50. The voltage of a fuel cell is a function of current density (output),

fuel utilization (fuel flow rate), cathode oxygen utilization, fuel cell temperature, and cathode carbon dioxide utilization. The fuel cell voltage can also change as a function of the amount of time the fuel cell operates. By correlating the fuel cell voltage to the known parameters and using the correlation to adjust at least one of the cathode gas flow rate, the fuel cell current (e.g., load), the amount of carbon dioxide which is allowed to cycle back to the cathode portion **51** of the fuel cell assembly **50** from the anode exhaust, the fuel cell assembly **50** can be prevented from over-utilizing carbon dioxide and maximize a carbon capture rate. FIG. **3** shows an exemplary relationship between carbon dioxide utilization of a fuel cell assembly and voltage across the fuel cell assembly. As shown in FIG. **3**, the fuel cell voltage varies according to the carbon dioxide utilization (e.g., the voltage varies inversely to the carbon dioxide utilization). The lines shown in FIG. **3** are lines of constant percent of carbon dioxide transferred, or utilized; specifically, FIG. **3** shows lines corresponding to 40%, 59%, 71%, 80%, and 92% of carbon dioxide transferred. The fuel cell voltage is a function of carbon dioxide utilization of the fuel cell, with the result that the fuel cell voltage of the fuel cell assembly is useable as an accurate measure of carbon dioxide utilization in the fuel cell. This relationship was determined through a series of controlled tests in a fuel cell. If the fuel cell design is modified, a new relationship may have to be determined based on additional tests with the new design.

(11) Referring to FIG. **2**, the controller **100** includes a memory **101** for storing programmable logic and data and a processor **102** for executing the programmable logic stored in the memory **101**. The controller is configured to receive a signal based on the measured voltage across the fuel cell assembly **50** from the fuel cell assembly voltage monitor **90**. Based at least on the measured voltage across the fuel cell assembly **50**, the controller **100** is also configured to determine an estimated carbon dioxide utilization of the fuel cell assembly **50** using the relationship between fuel cell voltage and carbon dioxide utilization described below.

(12) Specifically, the controller **100** may be configured to determine the estimated carbon dioxide utilization by determining an expected voltage across the fuel cell assembly **50** based on the average temperature of the fuel cell assembly **50** (as measured by one or more temperature sensors **95**, described below), a current density across the fuel cell assembly **50** (based on a measured fuel cell current as measured by a load controller **80**, described below), a fuel utilization of the fuel cell assembly **50**, and a cathode oxygen utilization of the fuel cell assembly **50**. The controller **100** may be configured to determine the estimated carbon dioxide utilization by comparing the measured voltage across the fuel cell assembly **50**, as measured by the fuel cell assembly voltage monitor **90**, to the determined expected voltage across the fuel cell assembly **50**. When the estimated carbon dioxide utilization is above a predetermined utilization threshold, the controller **100** is configured to reduce the carbon dioxide utilization of the fuel cell assembly **50** such that the carbon dioxide utilization of the fuel cell assembly **50** is at or below the predetermined high utilization threshold. Likewise, when the estimated carbon dioxide utilization is below a predetermined utilization threshold, the controller **100** is configured to increase the carbon dioxide utilization of the fuel cell assembly **50** such that the carbon dioxide utilization of the fuel cell assembly **50** is at or above the predetermined low utilization threshold.

(13) Still referring to FIG. **2**, the fuel cell system **1** also includes one or more temperature sensors **95** which are electronically coupled to the controller **100** and to the fuel cell assembly **50**. The temperature sensors **95** are configured to measure temperatures within the fuel cell assembly **50** and provide a signal to the controller **100** based on the measured temperatures in the fuel cell assembly **50**. The temperature sensors **95** may be located within the fuel cell assembly **50**. The temperatures as measured by the temperature sensors **95** is useable by the controller **100** in determining the average temperature of the fuel cell assembly **50**.

(14) Referring to both FIGS. **1** and **2**, the fuel cell system **1** also includes a load controller **80** configured to measure a current or load across the fuel cell assembly **50**. The load controller **80** is also configured to send a signal to the controller **100** based on the measured current across the fuel

cell assembly **50**. Based on the measured current as measured by the load controller **80**, the controller **100** is configured to determine the current density across the fuel cell assembly **50**. The load controller **80** is also configured to be controllable by the controller **100**. When the determined estimated carbon dioxide utilization is above a predetermined high threshold utilization, the controller **100** is configured to reduce the carbon dioxide utilization of the fuel cell assembly **50** by reducing the current across the fuel cell assembly **50**. Likewise, when the determined estimated carbon dioxide utilization is below a predetermined low threshold utilization, the controller **100** is configured to increase the carbon dioxide utilization of the fuel cell assembly **50** by increasing the current across the fuel cell assembly **50**.

(15) Still referring to FIGS. **1** and **2**, the flow transmitter **32** provides a measure of the flow rate of the fuel gas flow **30** to the controller **100**. The anode inlet gas flow **35** may also include additional fuel from the anode exhaust gas recycle portion **66''** as measured by the flow transmitter **74** and controlled by valve **73**. The controller **100** is configured to determine a fuel utilization of the fuel cell assembly based on the measured flow rate of the fuel gas flow **30** and the anode exhaust gas recycle flow **66''**.

(16) Still referring to FIGS. **1** and **2**, the fuel cell system **1** also optionally includes a hydrogen gas analyzer **60** which is configured to measure a hydrogen content of the anode exhaust gas flow **55** output from the anode portion **52** of the fuel cell assembly **50**. The hydrogen gas analyzer **60** is configured to provide a signal to the controller **100**. The controller **100** is configured to determine a fuel utilization of the fuel cell assembly **50** based on the measured hydrogen concentration of the anode exhaust gas flow **55** and thereby more accurately determine the fuel utilization of the fuel cell assembly **50**, as this relationship is more uncertain due to the addition of the anode exhaust gas recycle portion **66''** as measured by flow transmitter **74**. Using the hydrogen gas analyzer **60** allows for more optimal operation of the fuel cell assembly **50** by precision determination of fuel utilization allowing for fine adjustment of the fuel gas flow **30**.

(17) Still referring to FIGS. **1** and **2**, the fuel cell system **1** also includes a cathode inlet gas flow transmitter **26** which is configured to measure a flow rate of the cathode inlet gas flow **53** and provide a signal to the controller **100** based on the measured flow rate of the cathode inlet gas flow **53**. The fuel cell system **1** also includes a cathode inlet gas analyzer **27** which is coupled to the cathode portion **51** of the fuel cell assembly **50**. The cathode inlet gas analyzer **27** is configured to measure a composition of the cathode inlet gas flow **53**. Both the cathode inlet gas flow transmitter **26** and the cathode inlet gas analyzer **27** are configured to provide signals to the controller **100**. The controller **100** is also configured to determine a cathode oxygen utilization and carbon dioxide utilization based on the measured flow rate of the cathode inlet gas flow **53** and the composition of the cathode inlet gas flow **53**. The fuel cell system **1** also includes a cathode outlet gas flow transmitter **54a** which is configured to measure a flow rate of the cathode exhaust gas flow **54** and provide a signal to the controller **100** based on the measured flow rate of the cathode exhaust gas flow **54**. The fuel cell system **1** also includes a cathode outlet gas analyzer **54b** which is fluidly coupled to the cathode portion **51** of the fuel cell assembly **50**. The cathode outlet gas analyzer **54b** is configured to measure a composition of the cathode exhaust gas flow **54** and provide a signal based on the composition of the cathode exhaust gas flow **54** to the controller **100**. The controller **100** is configured to determine the cathode oxygen utilization and carbon dioxide utilization of the fuel cell assembly **50** based on the measured flow rate of the cathode exhaust gas flow **54** and the measured composition of the cathode exhaust gas flow **54**.

(18) Still referring to FIGS. **1** and **2**, the fuel cell system also includes a flue gas blower **10** which is configured to receive a flue gas flow **6** from an external flue gas generating device **5** (such as a carbonaceous-fuel-fired power plant, gas or oil-fired boiler, heaters, furnaces, or other types or forms of flue gas generating devices). The flue gas flow **6** includes carbon dioxide of an unknown concentration. A flow rate of the derivative gas flow **15**, which is derived from the flue gas flow **6**, from the flue gas blower **10** to an anode gas oxidizer **25** is controlled via the speed controller **11**. A

flow transmitter **12** is configured to measure a flow rate of the flue gas flow **15** and to send a signal to the controller **100** based on the measured flow rate of the flue gas flow **15**. When the determined estimated carbon dioxide utilization is above a predetermined threshold utilization, the controller **100** is configured to reduce the carbon dioxide utilization of the fuel cell assembly **50** by controlling the flue gas blower **10** to increase a flow rate of the flue gas flow **6** (or a derivative of the flue gas flow **6**, such as flue gas flow **15**) provided to the cathode portion **51** of the fuel cell assembly **50**.

(19) As shown in FIG. **1**, the fuel cell system **1** also includes an anode exhaust processor **65** which is fluidly coupled to the anode portion **52** of the fuel cell assembly **50**. The anode exhaust processor **65** is configured to receive the anode exhaust gas flow **55** from the anode portion **52** of the fuel cell assembly **50**. The anode exhaust processor **65** is also configured to provide an anode return gas flow **66**. The anode return gas flow **66** includes an anode exhaust gas return portion **66'** which is provided by the anode exhaust processor **65** via an anode exhaust gas return line to the anode gas oxidizer **25** whose exhaust is supplied to cathode portion **51** of the fuel cell assembly **50**. The flow of the anode exhaust gas return portion **66'** is controllable by the controller **100**, for example, using valve **71** based on a flow rate provided to the controller **100** by the flow transmitter **72**.

(20) The anode return gas flow **66** also includes an anode exhaust gas recycle portion **66''** which is provided by the anode exhaust processor **65** to the anode portion **52** of the fuel cell assembly **50** via an anode exhaust gas recycle line. The flow of the anode exhaust gas recycle portion **66''** is controllable by the controller **100**, for example, using valve **73** based on a flow rate provided to the controller **100** by the flow transmitter **74**. The anode exhaust processor **65** is also configured to provide a carbon dioxide product **67** using the carbon dioxide separated from the anode exhaust gas flow **55**. The flow of the carbon dioxide product **67** is controllable by the controller **100** using any suitable means (not shown). The anode exhaust processor **65** is also configured to provide a water product **68** using the water separated from the anode exhaust gas flow **55**. The flow of the water product **68** is controllable by the controller **100** using any suitable means (not shown). The anode exhaust processor **65** is also configured to provide a hydrogen product **69** using the hydrogen separated from the anode exhaust gas flow **55**. The flow of the hydrogen product **69** is controllable by the controller **100** using any suitable means (not shown).

(21) Using the valves **71**, **73** or by controlling the anode exhaust processor **65**, the controller **100** is configured to control the carbon dioxide utilization of the fuel cell assembly **50**. For example, the controller **100** can increase or reduce an amount of carbon dioxide product **67** provided by the anode exhaust processor **65** and thereby reduce the carbon dioxide utilization of the fuel cell assembly **50**. As another example, the controller **100** can reduce the amount of anode exhaust recycle portion **66''** and increase an amount of the anode exhaust gas return portion **66'** provided to the cathode portion **51** of the fuel cell assembly **50** and thereby reduce the carbon dioxide utilization of the fuel cell assembly **50**.

(22) As shown in FIG. **1**, the fuel cell system **1** also includes the anode gas oxidizer **25** which is coupled to the cathode portion **51** of the fuel cell assembly **50**. The anode gas oxidizer **25** is configured to receive a flow of flue gas **15** from the flue gas blower **10**, receive the anode exhaust gas return portion **66'** and receive an ambient air flow **23** from a fresh air blower **20** which receives an ambient air flow **7**. A speed controller **21** of the fresh air blower **20** controls a flow rate of the ambient air flow **7** into the anode gas oxidizer. A flow transmitter **22** measures the flow rate of the ambient air flow **23** and is configured to provide a signal to the controller **100** based on the measured flow rate of the ambient air flow **7**. By modulating the flow rate of the ambient air **23**, the amount of oxygen delivered to the fuel cell assembly **50** is controlled by the controller **100** to maintain a desired oxidization utilization.

(23) The anode gas oxidizer **25** is also configured to receive an oxidant gas, such as the ambient air flow **7** or flow **23**. The anode gas oxidizer **25** is also configured to provide the oxidized flue gas to the cathode portion **51** of the fuel cell assembly **50** as the cathode inlet gas flow **53**. The anode gas

oxidizer **25** is also configured to receive at least a portion of the anode exhaust gas return portion **66'** (described in more detail below). When the carbon capture process as described below is not implemented, all carbon dioxide present in the anode exhaust gas flow returns to the anode gas oxidizer **25** where it is subsequently provided to the cathode portion **51** of the fuel cell assembly **50**.

(24) To implement the carbon capture process, carbon dioxide from the flue gas flow **6** and/or derivative gas flow **15** is provided to the cathode portion **51** of the fuel cell assembly **50** by the anode gas oxidizer **25** via the cathode inlet gas flow **53**. Because the carbon dioxide content of the flue gas flow **15** is unknown, the carbon dioxide content of the cathode inlet gas flow **53** is also unknown. Once the carbon dioxide enters the cathode portion **51** of the fuel cell assembly **50**, the carbon dioxide is transferred to the anode portion **52** as carbonate ions by an electrochemical pumping (not shown). The carbon dioxide flux during this carbon dioxide transfer is determined by the load controller **80**. The transferred carbon dioxide available in the anode portion **52** is higher in concentration and easier to separate from the water and hydrogen in the Anode Exhaust Processor **65**. In the anode exhaust processor, carbon dioxide, water, and if necessary hydrogen are separated into different streams. If there is an insufficient concentration of carbon dioxide in the flue gas flow **6** or derivative gas flow **15**, then to avoid over-utilization of carbon dioxide by the fuel cell assembly **50**, at least one operating parameter of the fuel cell must be modulated. For example, the flue cell current (e.g., load) can be reduced, thereby reducing carbon dioxide flux. As another example, the amount of carbon dioxide product **67** can be reduced, thereby allowing more carbon dioxide exiting the anode portion **52** of the fuel cell assembly **50** to return to the cathode portion **51** via the anode gas oxidizer **25**. As another example, the amount of anode exhaust recycle portion **66''** can be reduce and the anode exhaust gas return portion **66'** can be increased to provide more carbon dioxide in the cathode inlet gas flow **53** that is supplied to the Fuel Cell Assembly **50**.

(25) According to another exemplary embodiment of the present disclosure, shown in FIG. **4**, a method **400** method for controlling a carbon dioxide utilization in a fuel cell assembly is shown. The method **400** includes the step **401** of measuring a voltage across the fuel cell assembly using a fuel cell assembly voltage monitor. The method **400** also includes the step **403** of determining an estimated carbon dioxide utilization of the fuel cell assembly based on at least the measured voltage across the fuel cell assembly by determining an expected voltage of the fuel cell assembly based on at least a temperature of the fuel cell assembly, a current density across the fuel cell assembly, and a fuel utilization of the fuel cell assembly, and determining the estimated carbon dioxide utilization based on a comparison between the measured voltage across the fuel cell assembly and the determined expected voltage across the fuel cell assembly. The method **400** also includes the step **405** of comparing the determined estimated carbon dioxide utilization of the fuel cell assembly to a predetermined threshold utilization. The method **400** also includes the step **407** of reducing the carbon dioxide utilization of the fuel cell assembly if the determined estimated carbon dioxide utilization of the fuel cell assembly is higher than the predetermined threshold utilization.

(26) Although the examples above refer to a single controller **100**, the aspects of the invention are not limited in this regard. In other examples, the system may include a plurality of controllers. For example, in some aspects, the system may include an anode inlet gas flow controller, a cathode inlet gas flow controller, an anode outlet gas flow controller, a cathode outlet gas flow controller, or a combination thereof.

(27) As utilized herein, the terms “approximately,” “about,” “substantially”, and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as

indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

(28) The terms “coupled,” “connected,” and the like, as used herein, mean the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members or the two members and any additional intermediate members being integrally formed as a single unitary body with one another or with the two members or the two members and any additional intermediate members being attached to one another.

(29) Although only a few embodiments of the present disclosure have been described in detail, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements, the position of elements may be reversed or otherwise varied, and the nature or number of discrete elements or positions may be altered or varied.

(30) Additionally, the word “exemplary” is used to mean serving as an example, instance, or illustration. Any embodiment or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs (and such term is not intended to connote that such embodiments are necessarily extraordinary or superlative examples). Rather, use of the word “exemplary” is intended to present concepts in a concrete manner. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the preferred and other exemplary embodiments without departing from the scope of the appended claims.

(31) Other substitutions, modifications, changes and omissions may also be made in the design, operating conditions and arrangement of the various exemplary embodiments without departing from the scope of the present disclosure. For example, any element (e.g., the anode exhaust hydrogen sensor, etc.) disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein. Also, for example, the order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. Any means-plus-function clause is intended to cover the structures described herein as performing the recited function and not only structural equivalents but also equivalent structures. Other substitutions, modifications, changes and omissions may be made in the design, operating configuration, and arrangement of the preferred and other exemplary embodiments without departing from the scope of the appended claims.

Claims

1. A method for controlling a carbon dioxide utilization in a fuel cell assembly, the method comprising: measuring a voltage across the fuel cell assembly using a fuel cell assembly voltage monitor; determining an estimated carbon dioxide utilization of the fuel cell assembly based on at least the measured voltage across the fuel cell assembly by determining an expected voltage of the fuel cell assembly based on at least a temperature of the fuel cell assembly, a current density across the fuel cell assembly, a fuel utilization of the fuel cell assembly, and a cathode oxygen utilization of the fuel cell assembly, wherein the estimated carbon dioxide utilization based on a comparison between the measured voltage across the fuel cell assembly and the determined expected voltage across the fuel cell assembly; comparing the determined estimated carbon dioxide utilization of the fuel cell assembly to a predetermined threshold utilization; and upon determining that the

- determined estimated carbon dioxide utilization of the fuel cell assembly is higher than the predetermined threshold utilization, reducing the carbon utilization of the fuel cell assembly.
2. The method of claim 1, wherein: an anode inlet gas of an anode portion of the fuel cell assembly comprises a fuel gas flow; and the method further comprises: measuring a flow rate of the fuel gas flow; and determining the fuel utilization of the fuel cell assembly based on at least the measured flow rate of the fuel gas flow.
 3. The method of claim 1, further comprising: measuring a hydrogen content of an anode exhaust gas of an anode portion of the fuel cell assembly; and determining the fuel utilization of the fuel cell assembly based on at least the measured hydrogen content of the anode exhaust gas.
 4. The method of claim 1, further comprising: measuring a flow rate of a cathode inlet gas of a cathode portion of the fuel cell assembly; measuring a composition of the cathode inlet gas; and determining the cathode oxygen utilization of the fuel cell assembly based on at least the measured flow rate of the cathode inlet gas and the composition of the cathode inlet gas.
 5. The method of claim 1, further comprising: measuring a flow rate of a cathode outlet gas of a cathode portion of the fuel cell assembly; measuring a composition of the cathode outlet gas; and determining the cathode oxygen utilization of the fuel cell assembly based on at least the measured flow rate of the cathode outlet gas and the composition of the cathode outlet gas.
 6. The method of claim 1, comprising: determining, by a controller, an average temperature of the fuel cell assembly based temperatures measured using a temperature sensor; wherein the temperature of the fuel cell assembly used to determine the expected voltage of the fuel cell assembly is the average temperature of the fuel cell assembly.
 7. The method of claim 1, comprising: determining the current density across the fuel cell assembly based on a current across the fuel cell assembly as measured by a load controller.
 8. The method of claim 1, further comprising: measuring a flow rate of a cathode inlet gas; measuring a composition of the cathode inlet gas; measuring a flow rate of a cathode outlet gas; measuring a composition of the cathode outlet gas; and determining the cathode oxygen utilization of the fuel cell assembly based on at least the flow rate of the cathode inlet gas, the composition of the cathode inlet gas, the flow rate of the cathode outlet gas, and the composition of the cathode outlet gas.
 9. The method of claim 1, wherein: the carbon dioxide utilization of the fuel cell assembly is reduced at least by reducing the current density across the fuel cell assembly.
 10. The method of claim 1, further comprising: upon determining that the determined estimated carbon dioxide utilization of the fuel cell assembly is below the predetermined threshold utilization, increasing the carbon utilization of the fuel cell assembly at least by increasing the current density across the fuel cell assembly.
 11. The method of claim 1, further comprising: upon determining that the determined estimated carbon dioxide of the fuel cell assembly is lower than the predetermined threshold utilization, increasing the carbon dioxide utilization of the fuel cell assembly.
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